Deven Desai

devendesai116@gmail.com | San Jose, CA | +1(669)-292-6556 | LinkedIn | GitHub

SUMMARY

Software Engineering Master's student and Stanford Hackathon winner with strong low level systems fundamentals, C++ fluency, and hands on experience building high performance, low latency retrieval architectures. Demonstrated track record optimizing data execution pipelines, reducing search latency by 35 percent, and tuning underlying data structures for high throughput computing.

PROJECTS

Low Latency Search Engine | High Performance Information Retrieval GitHub

- Queried and analyzed massive structured datasets in real time by engineering a custom high performance search interface as a solo developer using C++ and Java.
- Decreased execution latency by 35 percent by developing low level memory access routines, custom indexing strategies, and cache friendly data structures as measured by system performance benchmarks.
- Optimized processing throughput by 20 percent by implementing hardware performance tuning, profiling memory allocations, and eliminating bottleneck execution paths.

AgenticFlow | Concurrent Distributed Systems Engine GitHub

- Automated database schema transformations and complex code conversions across cloud systems by architecting a high throughput distributed service using Go and C++ concepts.
- Increased execution throughput by 40 percent by designing concurrent execution loops, gRPC microservice interfaces, and low overhead networking protocols as measured by backend execution traces.
- Maintained strict operational reliability across platform execution boundaries by implementing an automated differential privacy and verification layer.

OmniTrack AI | High Throughput Data Processing Platform GitHub

- Processed unstructured streaming inputs and high volume metadata natively in cloud infrastructure by engineering a zero to one distributed data processing pipeline using Python and C++.
- Achieved sub 100ms vector search latency by independently optimizing Qdrant retrieval algorithms for compute intensive, concurrent requests.
- Enhanced system stability and output correctness by 30 percent by building continuous execution monitoring and automated testing suites.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, Golang, SQL, TypeScript
Systems & Low Latency Engineering: Low Latency Systems, High Performance Computing, Memory Optimization
Networking & Infrastructure: gRPC, RESTful APIs, Network Protocols, UNIX Linux Systems Programming, Cloud Infrastructure
Algorithms & Performance: Data Structures & Algorithms, Object Oriented Design, Cache Optimization, System Architecture

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.